Week 10 web page: delivery guide

Instructor notes for Week10_ModelComparison.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week10_ModelComparison.html. Open it before class; the sliders need no internet once loaded.
- The page shows code and output already run. For live demos, run the matching section of `CODE_Week10_ModelComparison.R` instead (same seeds, same numbers).
- Zoom the browser to 125-150% for the projector; the sliders scale with it.
- Questions on the page are printed without answers. Answers are below; the *Land:* line is where discussion should end up.
- Use the floating table of contents to jump; it doubles as the outline.

Part 1: LRT, AIC, regularization (Monday, ~35 min)

Big picture (~4 min)

- Point at COMPARE in the workflow figure, then the three uses (null tests, selection, multimodel inference).
- Order to stress: validation and CV are the gold standard; LRT and AIC are single-fit shortcuts; ridge and lasso are a third route.

WOD: bias-variance tradeoff (~8 min)

- Show the WOD figure first; ask a few students for their 19% predictions.
- Then scroll to the 20-sample figure. Point at the right panel: both sets center near the truth (red), but the curve's spread is about 2.4 times the line's (SD 2.9 vs. 1.2 cm).
- If asked about bias: in this simulation the truth is a line, so the line is unbiased too. A constant model would show bias.

Likelihood ratio test (~8 min)

- Walk the four numbered steps, then the code: $D = 2.01$ vs. cutoff 7.81, $p = 0.57$.
- "Is the Poisson model nested in the negative binomial? Can we use an LRT here?" *Land:* yes, nested ($k \to \infty$), but that value is on the boundary of the parameter space, so the naive p is about twice too big. The deviance here is in the thousands, so it doesn't matter.

AIC (~10 min)

- "The no-wave model has weight 0.52. Is there a 52% chance it's the true model?" *Land:* no; relative support within this set (Bolker p. 284).
- `const_ak` is 1.48 AIC units behind with one useless parameter: "within 2" is not support (Arnold 2010).

Regularization (~12 min)

- Say up front the creeper data are simulated, so we know the truth.
- "Which covariates show up in nearly every top model?" *Land:* big trees, snags, road distance; the others are the best model plus one useless covariate, each about 2 units behind.
- Path plots: read the lasso right to left; big trees and snags enter first, then road distance, well before the rest.
- "The lasso estimates road distance at 0.07; the truth is 0.3. Is the lasso broken?" *Land:* no. Shrinkage trades bias for lower variance and better prediction. Use the lasso for prediction and covariate choice, not effect sizes (refit, or a shrinkage prior).
- 40 simulated data sets: AIC found the exact true set 12 times, lasso 13.
- **Behind?** Stop after the path plots; the CV demo can open the next lecture.

Discussion (~5 min)

- Q1 (northness) *Land:* report the uncertainty, not one list; big trees and snags are the safest claims.
- Q2 (grazing at 0) *Land:* zero means "not needed to predict counts here," not "no effect." Grazing may act through shrub cover or be too small to detect with 50 stations. A management question is causal: COMPARE part 3.

Part 2: prices, budgets, and CV (Lab 8 pre-lab)

Prices and budgets (~8 min)

- Order: price first (what `glmnet` computes), then the two price schedules, then the budget, then the bridge.
- Currency is fit (NLL units). Coefficient size buys lower NLL; the first bit of a useful coefficient buys the most.
- Price (λ): no spending limit; keep buying while the next unit lowers NLL by more than λ. $\lambda = 0$: size is free, MLE.
- Budget (t): fixed allowance, no price. Any t at least the MLE's size never binds, so it acts as unlimited.
- Bridge: fit at a price, total up what was bought: that's the matching t. Price for a given t = NLL one more unit of budget would buy. We tune λ; t pictures the spending.
- Schedules: lasso charges a flat λ per unit from the first unit; ridge's next unit costs $2\lambda\beta$, nearly free near 0. Let the slider question draw out what that means for zeros.

Lasso slider (`lasso_kink.html`)

- Start at $\lambda = 0$: dot at the MLE (0.4), both estimates equal.
- Move to 0.4: lasso estimate 0.2. Then to 1 and point at ridge: also 0.2. Different prices, same spending: both match a budget of $t = 0.2$.
- Ask where the lasso dot will land at 0.8 before moving there. It hits the kink at 0 and stays for every larger λ (try 1.5).
- Ridge keeps sinking but never reaches 0 (0.16 at $\lambda = 1.5$).
- "Why does the lasso buy nothing at all once $\lambda \geq 0.8$, while ridge always buys some?" *Land:* the first unit of β is worth 0.8 (the NLL's rate at 0). The lasso charges a flat λ for it, so at $\lambda \geq 0.8$ even the most valuable unit isn't worth the price. Ridge's first unit costs almost nothing ($2\lambda\beta$ near 0), so it's always worth buying.
- Don't draw the 2-D picture; Act 1 Part 2 asks for it.
- Glmnet tips and CV folds: read off the page; don't give away why standardize or whether to penalize β_0 (Act 2 1a-1b).

As taught

- Monday (Oct. 26) and the Lab 8 pre-lab (Oct. 27) covered Parts 1 and 2 as planned (reported Nov. 4).
- The ridge-as-prior board demo (Part 3, "Ridge is a prior") was not done; the instructor may run it before Lab 10 (Nov. 10).
- Wednesday (Oct. 28): Bayesian model comparison was **not covered**; it stays on the page as a bonus H1. The take-home WOD *Comparing Pond Models with LOO* was not handed out (moved to `wod/archive/`).
- The lecture included a sidebar on causal inference: how a DAG decides which variables to include and which to leave out. Added afterward to the page as "Causal diagrams" (`#dags`).

Causal diagrams (page section, added after class)

- Three panels: confounder (stand age; include), mediator (road distance -> big trees; leave out for the total effect), collider (bark insects; leave out).
- Collider sim (seed 1, 60 stations): big-tree slope 0.01 alone, 0.38 with insects in the model ($p = 0.001$); AIC prefers the collider model by 28 units.
- "Among stations with the same insect density, why would those with more big trees tend to have more creepers?" *Land:* with insects fixed, more big trees means insects should be high, so for them to be average, creepers must have eaten them down; big trees "explain away" the insect level, and the leftover points to more creepers.
- Q1 (lowest-AIC model, each slope as "the effect of"). *Land:* fair only if the chosen set blocks every confounding path and contains no mediators or colliders for that covariate; one model rarely does that for all 10 slopes at once. Need a DAG (subject knowledge), not more AIC.

- Q2 (prescribed burns; snags and canopy). *Land:* for the effect of burning, include neither (both are mediators). For the effect of snags, block the back-door path snags <- fire -> canopy -> creepers by including fire or canopy (either works).

Part 3: Bayesian comparison (bonus; not taught)

Why not AIC? (~3 min) No single $\hat{\theta}$; priors blur the parameter count.

Ridge is a prior (~6 min)

- Do the six steps on the board; the page has them for reference.
- Ridge-prior slider (`ridge_prior.html`):
  - Default $\tau = 0.05$: $\lambda = 200$, prior worth 9 routes at 0, shrinkage factor 0.47, estimate 0.031 (MLE 0.065).
  - Slide to 0.15: $\lambda = 22$, prior worth 1 route, factor 0.89, estimate 0.058. The posterior moves toward the likelihood.
  - Slide to 0.01: the prior dominates; estimate near 0.
  - Point out the posterior is narrower than both the prior and the likelihood (precisions add).
- "Which prior matches the lasso penalty?" *Land:* Laplace (double exponential), $\propto e^{-\lambda|\mu|}$: a spike at 0 and heavier tails.

WAIC and LOO (~12 min)

- Stan chunks take about a minute to run live; run them before class.
- "The full model is 2.1 elpd behind, SE 1.8. What can you conclude?" *Land:* about one SE; the data can't tell the models apart. Not the same as "no wave effect."
- **Behind?** Skip the DIC row.

Shrinkage priors (~12 min)

- Interval plot: northness (truth 0) under each prior; triangles are the glmnet lasso.
- "The lasso set northness to 0; the Laplace posterior says 0.08 (-0.07 to 0.28). Which is more honest?" *Land:* the posterior keeps the uncertainty a single zero hides; a Bayesian lasso never gives exact zeros.
- Horseshoe: sketch only.

Discussion (~5 min)

- Q1 (wave "no effect") *Land:* within about 1 SE is inconclusive; "not detected" is not "no effect"; corroborated, not proven.
- Q2 (northness) *Land:* for prediction either works; for inference, report the interval.
- Bonus (Bayes factors, RJMCMC): only if time remains.